

SAGE-IDOT

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1. Overall data

The dataset includes image and object-detection data collected from three SAGE sensor nodes:

- W06E: sampled on December 2, 2025.
- W042: sampled on August 9, 2025.
- W065: sampled on August 17, 2025.

The unit of analysis is a single timestamped image frame captured by a SAGE node. Each image may be associated with object-detection outputs generated by multiple YOLO models.

2. Core data components

The dataset is built from two primary data components:

Image data:

The provided sample dataset includes image folders for three selected node-date samples. Each sample folder contains the uploaded snapshot images associated with that specific node and day. Image filenames are based on the timestamp of the uploaded snapshot.

Object-detection data:

Detection results are stored in JSONL files. Each JSONL file corresponds to one selected SAGE node and date. The JSONL files contain timestamped records for that day, including image upload records and object-detection outputs.: **W06E_2025_12.jsonl**

The detection records include results from multiple object-detection models, including:

- YOLOv5n
- YOLOv8n
- YOLOv10n

For each model, the data may include detected object classes, confidence scores, bounding boxes, object counts by class, total detected objects, and inference time.

The repository also includes two example Jupyter notebooks in the **IDOT/jupyter_notebooks** folder. These notebooks demonstrate how to load the sample JSONL and image files, visualize

the provided YOLO detections, summarize detection counts over time, and run an additional off-the-shelf YOLO model on one sample image.

3. Image upload records

Images are identified using JSONL records where the field name is equal to: **upload**

These records contain the uploaded image URL in the value field. For example: **<https://storage.sagecontinuum.org/api/v1/data/.../1764719703877253000-snapshot.jpg>**

The image filename is extracted from this URL and matched to the corresponding local image file.

4. Variables

Each object-detection record contains a value field with model inference results stored as a JSON-formatted string. After parsing this field, the main variables are:

- **image_timestamp_ns**: Timestamp of the image associated with the detection record, stored in nanoseconds.
- **models_results**: A dictionary containing the detection results for each object-detection model. In this sample, the models include **YOLOv5n**, **YOLOv8n**, and **YOLOv10n**.

For each model, the following variables are included:

- **model**: Name of the object-detection model used.
- **detections**: A list of detected objects in the image. Each detected object includes:
 - **class**: The object category predicted by the model, such as car, person, boat, or traffic light.
 - **confidence**: The confidence score assigned to the detection by the model.
 - **bbox**: The bounding box coordinates for the detected object, stored as four numeric values: **[x_min, y_min, x_max, y_max]**
- **counts**: A summary of the number of detected objects by class for that model. For example: **{"car": 3, "person": 1}**
- **total_objects**: The total number of objects detected by that model in the image.
- **inference_time_seconds**: The time, in seconds, required by the model to process the image.

Each JSONL record also includes metadata fields describing the source of the data, such as the SAGE node, plugin, task, job, and zone. These fields include:

- **timestamp**: Timestamp of the JSONL record in UTC.

- **name:** Type of record, such as object.detections.all for detection outputs or upload for uploaded image records.
- **meta.vsn:** SAGE node identifier, such as W06E, W042, or W065.
- **meta.node:** Device/node identifier associated with the record.
- **meta.plugin:** Plugin that generated the record.
- **meta.task:** Task name associated with the record.
- **meta.zone:** Deployment zone associated with the record.
- **meta.filename:** Filename metadata. For upload records, this is usually snapshot.jpg.

For upload records, the value field contains the image URL. The image filename is extracted from the end of this URL. For example: **1764719703877253000-snapshot.jpg**

5. Output format

Each extracted sample is saved in a folder named using the SAGE node identifier and selected date: **<node>_<MM-DD-YYYY>/**

For example: **W06E_12-02-2025/**

Each folder contains two subfolders:

W06E_12-02-2025/

data/

W06E_2025_12_2025-12-02.jsonl

images/

1764698704479064838-snapshot

...

...

1764698704479064838-snapshot

The data/ folder contains the filtered JSONL records for that node and date. The images/ folder contains the uploaded snapshot images associated with the same node and date.

6. Included samples

The provided dataset contains the following three samples:

- W06E_12-02-2025: **snowy day sample**
- W042_08-09-2025: **sunny day sample**
- W065_08-17-2025: **rainy day sample**

Each sample includes both the JSONL data and the corresponding uploaded snapshot images for that day.

7. Questions to guide your exploratory analysis

The following questions are meant as starting points for exploration. Participants do not need to answer these exact questions; they are examples of research directions that could be pursued using the image data, detection outputs, and temporal structure of the samples.

1. Can the scene be reconstructed using only the object-detection outputs?

Using the detected classes, bounding boxes, confidence scores, and object counts, can participants describe or reconstruct what is happening in the image without first looking at the original photo? For example, can the detections alone suggest whether the scene contains a road, parked vehicles, pedestrians, or traffic infrastructure?

2. Can object detections be used to infer larger events happening in the scene?

Instead of studying individual detections, participants could investigate whether sequences of detections reveal higher-level events, such as traffic buildup, a train passing, a gate closing, vehicles stopping, or pedestrians entering a specific area. For example, if a railroad crossing is present, can the movement or disappearance of vehicles, changes in detected objects, or the position of a gate-like object help infer whether a train is passing?

3. How can model disagreement be used as a signal rather than treated only as an error?

When YOLOv5n, YOLOv8n, and YOLOv10n disagree, what does that disagreement reveal about the image? Participants could examine whether disagreement is associated with weather, lighting, occlusion, small objects, reflections, motion blur, or ambiguous objects. Model disagreement may help identify difficult visual conditions or moments where human review is most needed.

4. Can weather conditions change not only detection accuracy, but the interpretation of the scene?

Snow, rain, and sunlight may affect visibility, contrast, object boundaries, and reflections. Participants could investigate whether the same types of objects are detected differently across

weather conditions, and whether weather causes certain objects to become over-detected, under-detected, or confused with other categories.

5. Can temporal patterns reveal routines or anomalies in the environment?

By aggregating detections over time, participants could explore whether the scene has recognizable daily rhythms, such as commuting peaks, low-activity periods, or sudden bursts of activity. They could also look for unusual moments where detection counts sharply increase or decrease and inspect the corresponding images to understand what happened.